In [27]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [12]:
```
df=pd.read_csv('sales_data - Sheet1.csv')
print(df.head())
```

```
         Date    Product  Sales  Quantity Region
0  01-01-2023  Product A    200         4  North
1  02-01-2023  Product B    150         3  South
2  03-01-2023  Product A    220         5  North
3  04-01-2023  Product C    300         6   East
4  05-01-2023  Product B    180         4   West
```

In [17]:
```
print(df.isnull().sum())
```

```
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
```

In [18]:
```
df['Sales']=df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
```

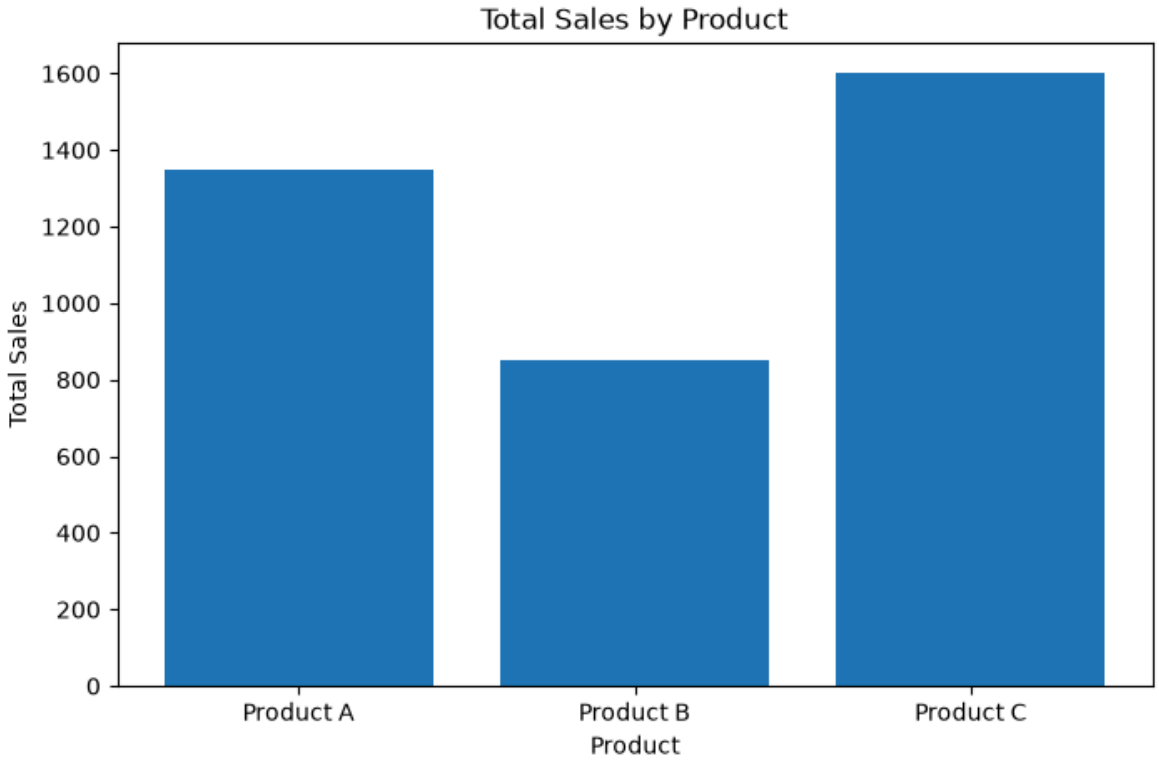
In [19]:
```
print(df.describe())
```

```
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
```

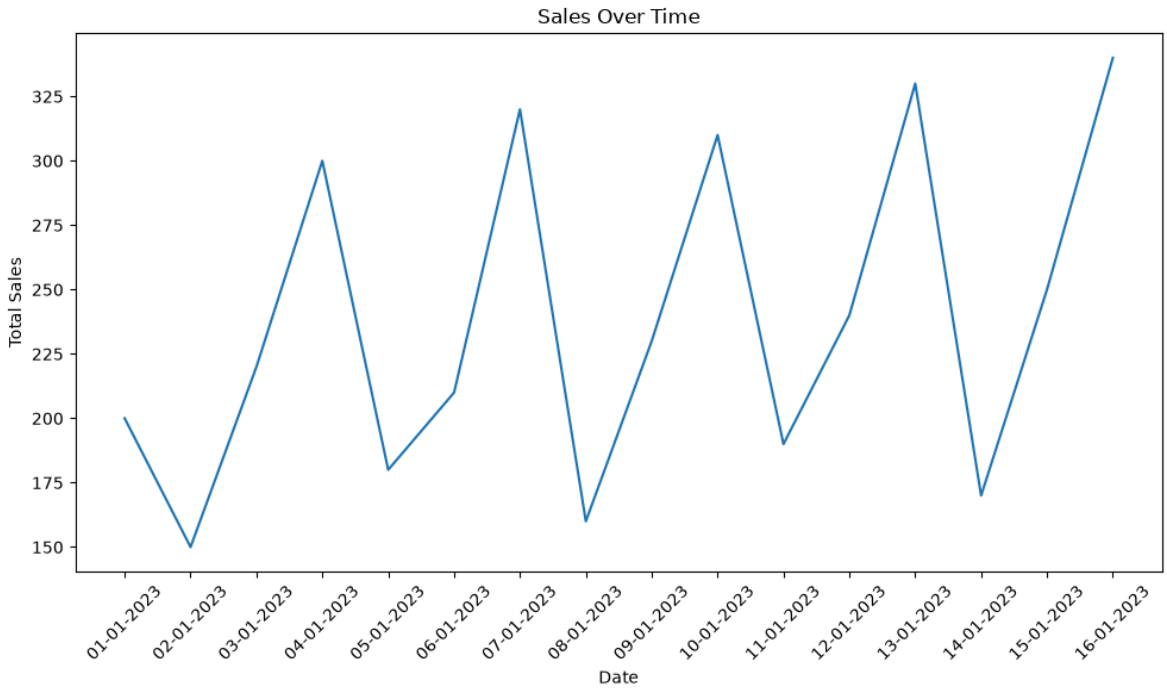
In [20]:
```
product_summary=df.groupby('Product').agg({'Sales': 'sum','Quantity': 'sum'}).re
print(product_summary)
```

```
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```

In [21]:
```
plt.figure(figsize=(8,5))
plt.bar(product_summary['Product'],product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
```



In [33]:
```
df['Data']=pd.to_datetime(df['Date'],format='%d-%m-%Y')
sales_over_time=df.groupby('Date').agg({'Sales':'sum'}).reset_index()
plt.figure(figsize=(10,6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In [34]:
```
pivot_table=df.pivot_table(
    values='Sales',
    index='Region',
```

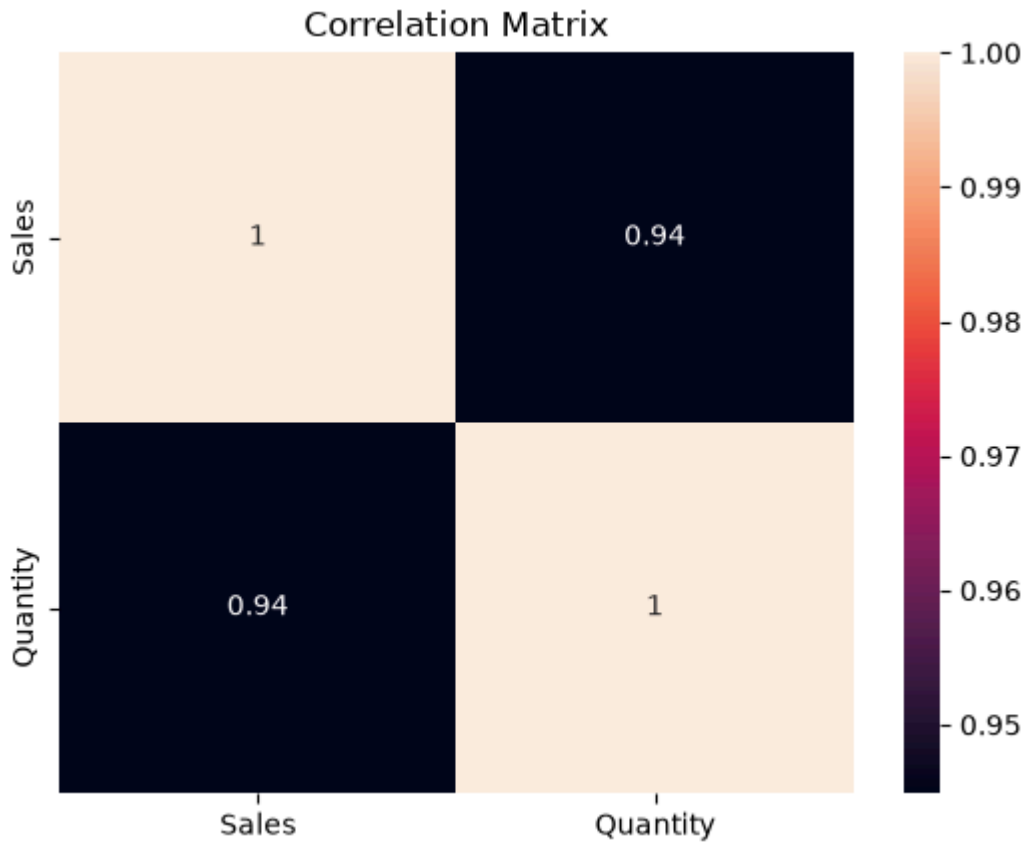
```
    columns='Product',
    aggfunc=np.sum,
    fill_value=0)
print(pivot_table)
```

```
Product  Product A  Product B  Product C
Region
East             0          0       1600
North         1350          0          0
South            0        480          0
West             0        370          0
```

In [37]:
```
correlation=df[['Sales','Quantity']].corr()
print(correlation)
```

```
              Sales  Quantity
Sales     1.000000  0.944922
Quantity  0.944922  1.000000
```

In [38]:
```
sns.heatmap(correlation, annot=True)
plt.title('Correlation Matrix')
plt.show()
```



In [ ]: